

(100)

PG 1

① GIVEN:

IN RX FOR 500 DAYS @ 625K

$W_{Ti}: 600 \mu m$

$f_{H_2}: 14\% \rightarrow 0.14$

$P_{BZ}: 1.56$

$P_{Zr}: 6.5 g/cc$

$P_{Zr}: 5.68 g/cc$

18/18

$$a.) d^* = 5.1e^{-550/T} = 5.1e^{(-550/625)} = 2.115 \mu m$$

$$t^* = 6.62 \times 10^{-7} e^{11949/K} = 6.62 \times 10^{-7} e^{(11949/625)} = 133.01 d$$

500 > 133.01 d $\Rightarrow$ POST-TRANSITION REGIME

$$K_L = 7.48 \times 10^6 e^{-12500/T} = 7.48 \times 10^6 e^{(-12500/625)} = 0.0154 \mu m/d$$

$$d = d^* + K_L(t - t^*) = 2.115 \mu m + 0.0154 \mu m/d (500 - 133.01 d)$$

$$d = 7.77 \mu m$$

$$b.) C_{H_2}^{CLAD} \text{ WT. PPM} = \frac{2f \times d \times P_{Oxide} \times f_{ZrO_2} \times \frac{M_H}{M_O}}{(1 - \frac{f}{P_{BZ}}) \times P_{Zr} \times K_L} \times 10^6$$

$$f_{ZrO_2} = \frac{32}{91+32} = 0.2602$$

$$\frac{M_H}{M_O} = \frac{1}{16}$$

$$W = 14.7 d = 7.77 \cdot 14.7 = 114.2$$

$$114.2 \text{ mg/dm}^2 = 1.142 \times 10^{-3} \text{ g/cm}^2$$

$$M_H = 0.14 \left(\frac{2(1.142 \times 10^{-3})}{16} \right)$$

$$M_H = 1.999 \times 10^{-5} \text{ g/cm}^2$$

$$C_{H_2}^{CLAD} \text{ WT. PPM} = \frac{2(0.14) \times (7.77) \times (5.68) \times (0.2602) \times (\frac{1}{16})}{(600 - \frac{7.77}{1.56}) \times (6.5)} \times 10^6$$

$$C_{H_2}^{CLAD} = 51.96 \text{ WT. PPM}$$

Equations for NE533

$$\frac{\delta^n}{2} = K^n t \rightarrow \delta = K t^{1/n}$$

$n \approx 4$ for Zr alloys

$$\delta^*(\mu m) = 5.1 \exp\left(\frac{-550}{T}\right)$$

$$t^*(d) = 6.62 \times 10^{-7} \exp\left(\frac{11949}{T}\right)$$

$$K_L\left(\frac{\mu m}{d}\right) = 7.48 \times 10^6 \exp\left(\frac{-12500}{T}\right)$$

$$\delta(\mu m) = \delta^* + K_L(t - t^*)$$

$$w = (m_2 - m_1)/S \quad (mg/dm^2)$$

$$\delta(\mu m) = \frac{w}{14.7}$$

$$C_H^{clad}(wt.ppm) = \frac{2f \times \delta \times \rho_{oxide} \times f_{ZrO_2}^O \times \frac{M_H}{M_O}}{(t - \frac{\delta}{PBR}) \times \rho_{metal}} \times 10^6$$

PG 2

2/2 ② FOR AGG. CORROSION OF Zr CLADDING, THE LIMITING STEP FOR RATE OF CORROSION IS THE TRANSPORT OF OXYGEN SPECIES THROUGH THE OXIDE LAYER.

8/8 ③ HYDRIDES FORM IN CLADDING @ COOLER AREAS WITH TENSILE STRESS BECAUSE H₂ MIGRATES DOWN A TEMP. GRADIENT VIA THE SORBT EFFECT. IN ORDER FOR A HYDRIDE TO START, ZrO₂ REACTIONS MUST OCCUR TO ENTER CLADDING AT A HIGH SOLUBILITY RATE.

12/12 ④ A RIA IS A REACTIVITY INITIATED ACCIDENT IN WHICH POSITIVE REACTIVITY OCCURS CAUSING PRX ↑ QUICKLY ALONG WITH T_q OF THE FUEL. IN A PWR, THIS WOULD BE A CONTROL ROD EJECTION OR UNCONTROLLED OUTWARD ROD MOTION. IN A BWR THE RIA IS FROM A CONTROL ROD DROP EVENT RESULTING IN CTRL BLADES BEING DROPPED OUT OF THE CORE.

DURING A RIA, PRX + TEMP ↑↑ SIGNIFICANTLY RESULTING IN FUEL EXPANSION RAPIDLY, FISSION GAS RELEASE RESULTING IN FUEL ROD INTERNAL PRESSURE ↑ ALL OF WHICH INCREASES STRAIN ON THE CLADDING. IF CHF IS EXCEEDED, DNB OR DRYOUT OCCURS ⇒ CLADDING TEMP RAPIDLY ↑↑ ⇒ CLADDING MELT OR BURST OCCURS.

12/12 ⑤ A LOCA IS A LOSS OF COOLANT ACCIDENT IN WHICH COOLANT FLOW IS LOST, USUALLY FROM A MECHANICAL SYSTEM FAILURE CREATING A COOLANT LEAK. DURING A LOCA, THE LOSS OF COOLANT PREVENTS REMOVING DECAY HEAT FROM THE FUEL RODS OR CELLS ⇒ FUEL ROD TEMPS BEGIN TO RISE. THIS RESULTS IN INCREASED CLADDING OXIDATION + HYDRIDING AND HIGH CLADDING TEMPS WHICH LEAD TO CLADDING FAILURE.

IN A RIA THE ACCIDENT IS RAPID, WHEREAS A LOCA IS A SLOW INCREASE IN FUEL ROD/CELL TEMPERATURES.

PG 3

⑥ BURNUP WILL ↑ ACCIDENT VULNERABILITY SINCE IRRADIATED FUEL + CLADDING WILL FAIL @ LOWER POINTS COMPARED TO NEW FUEL ROD ASSEMBLIES. AS BURNUP OCCURS, FISSION PRODUCT GAS WILL ↑ INSIDE FUEL ROD, FUEL CONDUCTIVITY ↓ AND CORROSION ON THE CLADDING WILL ↑. AS OXIDE LAYERS ↑, H_2 PICKUP ↑ ⇒ HYDRIDING ↑ + CLADDING DUCTILITY ↓. THEREFORE BURNUP WILL ↑ THE PROBABILITY OF AN ACCIDENT.

⑦ 4 PATHWAYS TO MAKE FUEL/CLADDING ACCIDENT TOLERANT:

- a. IMPROVE CLADDING THROUGH ↑ SHOCK RESISTANCE + MELTING POINTS.
- b. IMPROVE THE FUEL BY ↑ MELTING MARGIN, ↓ FUEL DISPERSION, LOWER REQUIRED OPERATING TEMPS
- c. IMPROVE FISSION PROD RETENTION BY RETAINING GASEOUS, SOLID OR LIQ FISSION PRODUCTS MORE EFFECTIVELY
- d. IMPROVE STEAM REACTION EFFECT BY ↓ OXIDATION RATES, ↓ H_2 GENERATION + HYDRIDE EMBRITTLEMENT POTENTIAL.

⑧ A NEAR TERM ATF CONCEPT IS CHROMIUM Zr ALLOY CLADDING BECAUSE IT IS EASILY SWAPPED W/ EXISTING FUEL RODS/CELLS, AND ↓ OXIDATION AT HIGH TEMPERATURES SIGNIFICANTLY. LESS OXIDATION → LESS H_2 → LESS HYDRIDE EMBRITTLEMENT.

⑨ IN A HIGH TEMP STEAM ENVIRONMENT, Zr CLADDING WILL REACT AS $Zr + 2H_2O \rightarrow ZrO_2 + 2H_2$. THIS REACTING INCREASES SIGNIFICANTLY ABOVE 1200°C IN WHICH THE Zr WILL TRANSITION FROM AN α -Zr TO A β -Zr WHICH WILL CAUSE A BRITTLE FRACTURE BEHAVIOR ON THE CLADDING.

⑩ THREE LIMITING PHENOMENA IN LWRs ARE:

- a. CLADDING OXIDATION + H_2 PICKUP. OXIDE ↑ ⇒ H_2 ↑ ⇒ EMBRITTLEMENT
- b. DEPARTURE FROM NUCLEATE BOILING, IF CHF IS EXCEEDED, STEAM WILL BLANKET THE HEAT TRANSFER SURFACE.
- c. PELLET-CLADDING INTERFACING - FUEL EXPANSION INTO CLADDING ↑ HOOP STRESS WHICH ↑ SCC EFFECTS.

PG 4

- 6/6
- AND SAFETY/
- ⑪ CRUD EFFECTS FUEL PERFORMANCE[↑] BY LOWERING HEAT TRANSFER FROM FUEL ROD TO COOLANT WHICH $\uparrow$ FUEL TEMPS. IT CAN ALSO IMPACT COOLANT FLOW $\Rightarrow$ EFFECTING HEAT TRANSFER CHARACTERISTICS.

CRUD EFFECTS SAFETY/ IN THAT IT CAN DEPOSIT OUTSIDE THE CORE $\uparrow$ RADIATION LEVEL TO PERSONNEL.

- 6/6
- ⑫ TWO WATER CONTROLS IN LWRs ARE:
- PH CTRL USING BORIC ACID + LiOH - BORIC ACID AS A REACTIVITY CTRL MECHANISM + LiOH AS A PH MAINTAINER. BY CTRL'NG PH PROPERLY, CRUD IS $\downarrow$ IN THE CORE.
 - IN BWRs NOBLE METAL INJECTION - THE NOBLE ~~METALS~~ METALS ARE USED TO IMPROVE H_2 CHEMISTRY TO $\downarrow$ ELECTROCHEMICAL CORROSION AND MINIMIZE INTERGRANULAR SCC.

- 8/8
- ⑬ TWO MICROSTRUCTURE PHENOMENA IN MOX:
- FUEL STRUCTURE & VOID FORMATION - AT HIGH TEMPS, THE FUEL PORES MOVE TO THE FUEL CENTER WHICH CREATES NEW GRAIN STRUCTURES WHICH MOVE OUTWARD TO THE MOST OUTER REGION.
 - REDISTRIBUTION - PLUTONIUM & OXYGEN MOVE RADIALLY DURING RX OPERATION. THIS CAN CAUSE AN ENRICHMENT EFFECT OF PU NEAR THE FUEL CENTER.
 - SOG - FORMED AS CsMoO_4 , IT IS A LAYER WHICH FORMS BETWEEN THE FUEL + CLADDING + $\downarrow$ FUEL SWELLING @ HIGH TEMPS.

